CAMPERMINDER
ENHANCE YOUR JOURNEY

CAMPERMINDER

# LEVEL

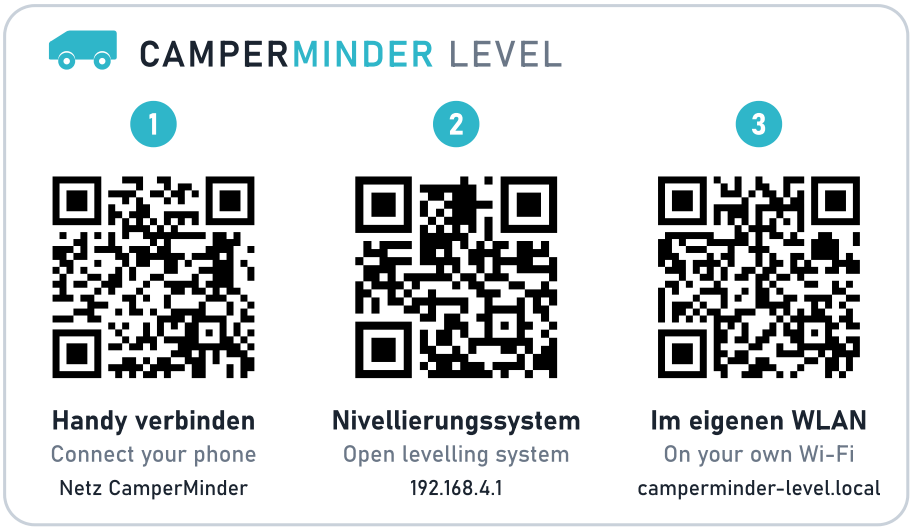
Manual – from unboxing to the controls on your phone

**No app, no account, no pairing.** Everything that goes beyond the first minute is an optional extra that you add when you need it.


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As of: September 2026

# 1 · Ready in one minute




The sticker on the device. The codes in this manual work exactly like the ones on the device.

This sticker with three codes is on LEVEL. You do not need to type anything: for the first start, scan **code 1 and then code 2**. Code 3 only comes into play later, once the device is on your Wi-Fi.

**What to scan with:** the camera app of your phone – hold the camera on the code until a notice appears, and tap it. On Android, Google Lens works as well. If the camera reads nothing, scanning of QR codes may be switched off in its settings.




The whole first start.

1. **Connect power.** LEVEL runs on **5 to 36 volts** and has two connections for this:
   - **USB-C**, 5 V: to a USB socket in the vehicle, a USB power supply or a power bank.
   - **Screw terminal**, 7 to 36 V: **VIN** to plus, **GND** to minus of the leisure battery, i.e. straight onto the 12 or 24 volt on-board supply. Plus and minus the wrong way round? No problem: LEVEL is protected against wrong connection, nothing gets damaged.

   Both may be plugged in at the same time; the stronger source then supplies the power. The 🟢 **green** light shows that power is on. After a few seconds the Wi-Fi network **CamperMinder** is ready.
2. **Scan code 1: connect your phone.** The phone asks whether it should join the Wi-Fi network **CamperMinder**: confirm **Connect** or **Join**. LEVEL sends out this Wi-Fi itself; it is not your own Wi-Fi, such as the router in the vehicle. It needs no password from the factory.

   Because this Wi-Fi has no internet, every phone reacts a little differently:

- **iPhone:** Often the control page now opens **by itself**, in a window called *Log In*. Then you are done and do not need code 2. If you close the window, the iPhone asks how to continue: **Use Without Internet.** If nothing opens, carry on with code 2.
- **Android:** The control page usually does **not** open by itself. The phone may point out that the Wi-Fi has no internet and ask whether to stay connected: **Yes** or **Stay connected.** Then carry on with code 2.

3. **Scan code 2: open the control page.** The page `192.168.4.1` opens in the browser, and **LEVEL** is ready.

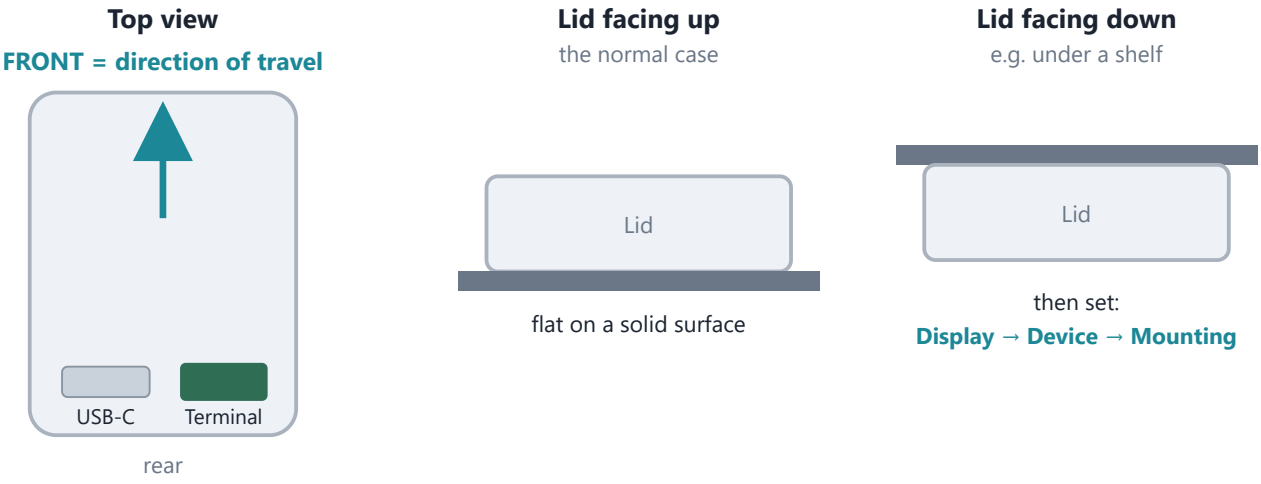
Nothing more is needed for operation. **LEVEL** needs no internet, and you can open the page again at any time in the same way: scan code 2 while on the CamperMinder Wi-Fi.

### Without the sticker, or if a code does not work

- **Instead of code 1:** In the Wi-Fi settings of your phone choose **CamperMinder**. From the factory this Wi-Fi has no password. If you have set one of your own under *System → Own network*, yours applies – then code 1 no longer fits, and you have to choose the network by hand.
- **Instead of code 2:** Type `192.168.4.1` into the browser. Some browsers turn that into a search; then type the full `http://192.168.4.1`.
- **Shortcut on Samsung phones:** *Settings → Connections → Wi-Fi*, tap the gear next to the connected Wi-Fi **CamperMinder**, then **Manage router**. That also opens `192.168.4.1`.

> **An icon like an app.** Open the page for this in the right browser, not in the *Log In* window of the iPhone: on the iPhone call up `192.168.4.1` in Safari, then *Share → Add to Home Screen*; on Android in Chrome, then *⋮ → Add to Home screen*. Code 2 already opens the page in the right browser. The icon opens it full screen – where other devices need an app, **LEVEL** has the controls built in. New features arrive later as an update to the device; you never have to update an app.

## 2 · Installation



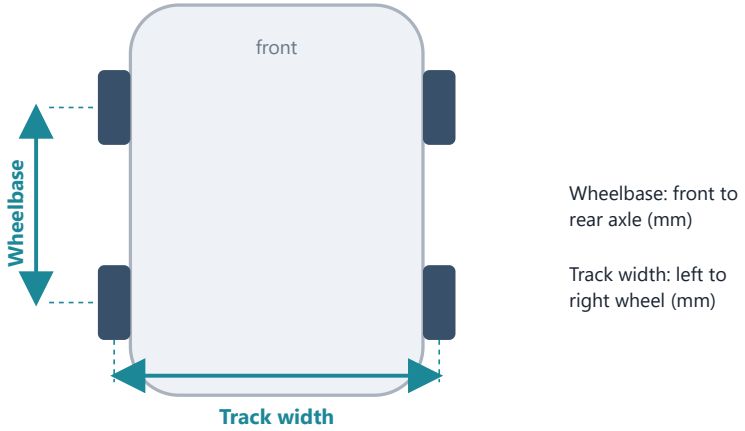

The arrow on the housing wall points forwards. The USB socket and the terminal are at the rear.

- **The arrow points forwards.** It is on the housing wall and points in the direction of travel; the USB socket and terminal are at the rear. If the device is installed turned by 90 degrees, it mixes up lengthways and crossways.
- **Lid facing up** is the normal case. If you stick the housing under a shelf, set *Display → Device → Mounting* to **Lid facing down** and then calibrate again (see section 3).
- **Firm and flat.** Screw or stick it onto a level, solid surface. A device that wobbles measures the wobble.

## 3 · Setting up, once

### Step 1: Enter your vehicle

*Display → Vehicle*: vehicle type, wheelbase and track width. From these dimensions **LEVEL** works out how many centimetres each corner has to be raised.




The two dimensions. Accurate to ±5 cm is perfectly fine – guessed made-up values are not.

The dimensions determine two things: how high each corner has to be raised – and when “level” applies. Grossly wrong dimensions therefore also shift the point at which the device says “level, stop”.

### Step 2: Set the vehicle level and calibrate

Set the vehicle up **truly level** once, checked lengthways and crossways with a spirit level, and tap **Calibrate level** on *Display*, below the views. The sensor is already calibrated from the factory; this step compensates for how the device sits in your vehicle.

- **This is the biggest source of error, not the sensor.** Whatever is off here stays permanently baked into all later centimetre figures.
- While pressing, **do not move about in the vehicle**.
- Afterwards **leave it powered for about a minute**: only then is the zero point stored permanently and survives every restart.
- You have to **calibrate again** after every change of the mounting (up/down) and whenever you move the device.

**How long does the calibration last?** As long as the device is not moved, indefinitely – it survives power cuts and software updates. A motorhome, however, goes through a 40 to 50 degree difference between a winter night and a summer midday, and the zero point of every acceleration sensor drifts a little with the temperature. If the display seems to drift over the seasons, simply calibrate again.

## 4 · Understanding the display

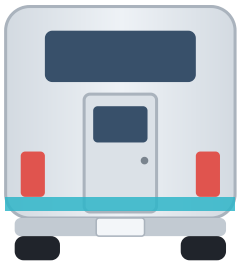
The control page shows your vehicle in three views: from above with a moving bubble, from the side and from behind, plus an instruction in plain language saying which corner has to be raised by how many centimetres. The graphic follows the vehicle type you have set.



Motorhome: top view



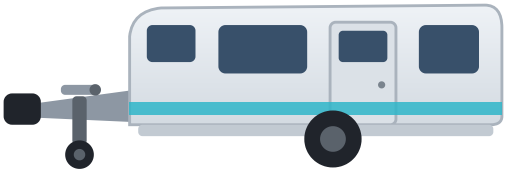
Side view (along)



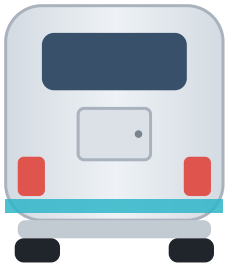
Rear view (across)



Caravan: top view



Side view (along)



Rear view (across)

### The tolerance is the yardstick

The tolerance is not just the limit between “level” and “not level”, but the yardstick of the display itself: **tolerance limit = edge of the green zone.** If the vehicle is within the tolerance, the bubble sits in the middle, the readouts sit in the middle and the side views are horizontal.

That is why the display looks coarser with a 5 cm tolerance than with 1 cm. It shows the same measurement, only on the scale you have specified: someone who allows 5 cm does not want to see at 4 cm that 4 cm are missing – they want to see that they are done.

**Once “level”, always “level”,** until the tilt clearly goes beyond the tolerance. That way the display does not flip back and forth at the limit.

### How calm should the display be?

That is a matter of taste: one person wants it to stand as if nailed down, another wants to see every twitch. That is why there are two controls, on the control page under **Display**. They take effect immediately, without a restart – for trying out in the vehicle.

| What you observe | Control | Direction |
|---|---|---|
| still jitters when parked | **Display steadiness** | higher |
| lags behind when driving up | **Display steadiness** | lower |
| follows too jerkily when manoeuvring | **Display steadiness** | higher |
| flips back and forth at the tolerance limit | **Hold range** | higher |

**Display steadiness** runs from 0 to 10, default 5. It changes nothing about the measurement and nothing about the tolerance – only the patience.

| Steadiness | Response time | What you notice |
|---|---|---|
| 0 | immediate | the number is alive, every twitch shows at once |
| 2 | about 1 s | slight wandering of the last digit |
| **5** | about 6 s | **stands still when parked** |
| 10 | about 24 s | stands just as still, but follows more sluggishly |

> **To try it out, turn it down, not up.** Above about 3 the display no longer becomes visibly calmer, it just reacts later. You do not see the difference between 5 and 10 when parked, but when you lean against the vehicle.

**Hold range** in per cent, default 125: how far the tilt may go beyond the tolerance before "level" is taken back. 100 % means immediately – then the display flips back and forth at the limit.

### Precision mode

If the display is too coarse for you, switch on **Precision mode**: there the bubble sits at its true position even within the tolerance, the angles are shown with two decimal places, and instead of centimetres **Precise tolerance** applies, in degrees – the same number for both axes. The switch is on the control page under *Vehicle*.

## 5 · What the lights say

| Light | Meaning |
|---|---|
| 🟢 **green**, on | Power is on |
| 🔴 **red**, long on, long off | LEVEL's own Wi-Fi is open: code 1, then code 2 – or connect to **CamperMinder** by hand and open `192.168.4.1` |

| | |
|---|---|
| 🔴 **red**, one short flash every three seconds | The device is on your Wi-Fi |
| 🔴 **red**, steady at half-second intervals | The sensor is not responding. Please report it. |
| 🔴 **red**, frantic double flashing, plus three tones | **Guard alarm:** The vehicle has changed its position |
| ○ **red** off | The light has been switched off with the *Status LED* switch. An alarm still flashes. |

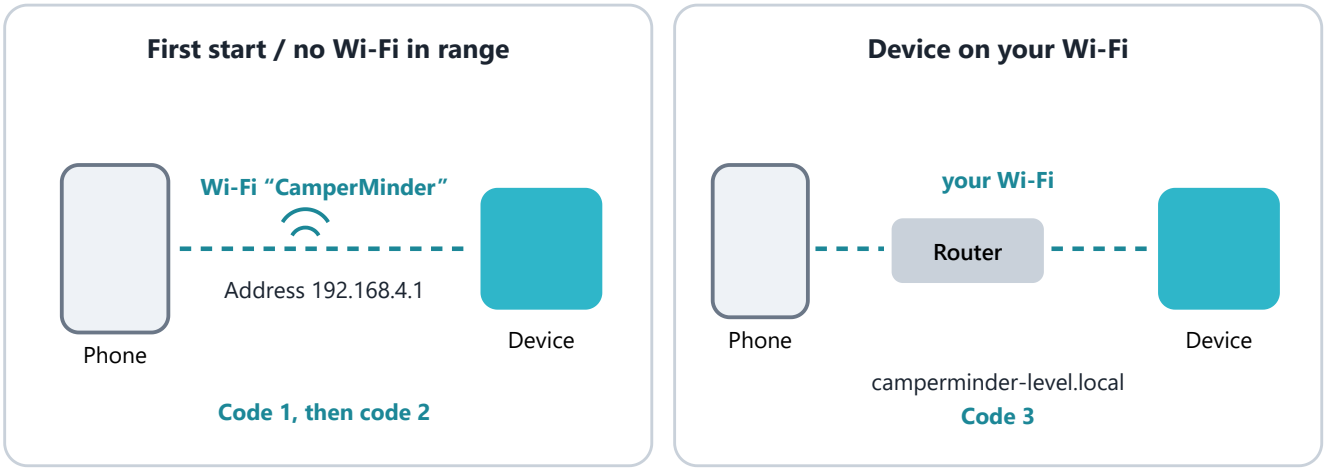
## 6 · Language

The control page speaks **German and English**. Without a choice of your own it follows the language setting of the phone; change it under the tab **System** → **Language**.

The choice applies **per phone**, not per device – two people in one vehicle can each read in their own language. The names of the readings stay German, so that Home Assistant and MQTT keep matching up when the language is changed.

## 7 · Joining your own Wi-Fi (optional extra)

Only needed for automatic updates and Home Assistant. Without this step everything else works unchanged, on the road anyway.




Two ways to the same device. If your Wi-Fi is out of range, the device opens its own network again after about 20 seconds.

1. Connect the phone to the Wi-Fi network **CamperMinder** (code 1) and open the page with **code 2** – code 2 opens it in the right browser. **Do not work in the *Log In* window** of the iPhone: it closes as soon as the phone switches networks, and then the page can no longer search for the device. There go to *System* → *Wi-Fi*, enter the name and password of your Wi-Fi, and tap **Save and connect**.
2. **Stay on the page.** The device restarts and logs on to your Wi-Fi. The CamperMinder Wi-Fi disappears, and your phone returns to your Wi-Fi by itself. The page then searches for the device there and, usually

after less than a minute, shows: *Found: the device is reachable on your Wi-Fi at 192.168.178.45* (your number will be a different one).

3. Tap **Open on home network** and add the icon to your home screen again from there.

That gives you two icons, and both stay correct. If the device is on your Wi-Fi, the new address applies. If your Wi-Fi is not in range, the device opens its own network again after about 20 seconds, and `192.168.4.1` applies. The credentials stay saved.

**Code 3 on the sticker** opens the device on your Wi-Fi by its name, `camperminder-level.local`, no matter which address your router gives it. For this your phone must be on your Wi-Fi, not on the CamperMinder Wi-Fi. On the iPhone this works, on Android not on every phone. If code 3 opens nothing there, use the icon you created in step 3.

| You are … | Code |
|---|---|
| at the first start, or your Wi-Fi is not in range: the device has its own network open | **1**, then **2** |
| the device is on your Wi-Fi | **3** |

**If the CamperMinder Wi-Fi comes back after half a minute,** logging on did not work, usually because of a typo in the password. The page says so as well. Simply connect again, correct it, save.

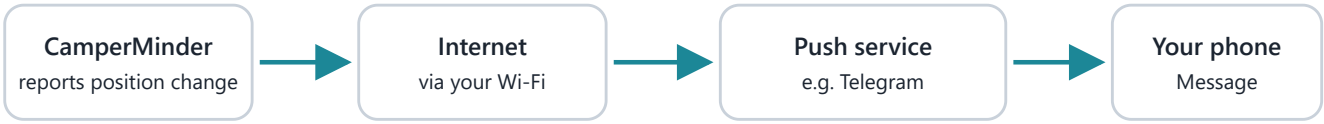
### If the page does not find the device

- **Look in the router.** In the device list of your router the device appears under its name, usually **camperminder-level**, on a FRITZ!Box under *Home Network → Network*. Open the address next to it in the browser. The page gives the exact name under *System* as *Network name*.
- **Scan code 3** or type `camperminder-level.local` into the browser. On the iPhone this almost always works, on Android not on every phone.
- **Look it up next time.** The device remembers its address. The next time you are on the CamperMinder Wi-Fi, it is shown at the top under *System → Wi-Fi*.

The search finds nothing in guest Wi-Fis that isolate their devices from each other, and in unusually set-up networks. Then all that is left is a look in the router.

## 8 · Alarm on your phone: setting up the remote alert (optional extra)

The guard reports a change of position to everyone who can reach the device – on its own network. If you are on the beach while the vehicle is being jacked up, you only find out when you get back. That is exactly what the remote alert is for: the device sends the message itself to a push service that **you** run or use.




The route of the message. No account with us, no data with us.

> **No account with us, no data with us, no availability guarantee.** That is the price for it costing nothing and nobody reading along.

Tab **System** → section **Remote alert**: enter the address, save, **Send a test message**. The last step is an absolute must – an alert path nobody has tried is not an alert path.

## What goes into the field

Two modes, told apart by the placeholder `{text}`:

| | Address | What the device does |
|---|---|---|
| **A** | without `{text}` | sends the message as the body of a POST request |
| **B** | with `{text}` | inserts the message, encoded, into the address and calls it up |

## Services

| Service | Example | Mode | Cost |
|---|---|---|---|
| **Telegram** | `https://api.telegram.org/bot<TOKEN>/sendMessage?chat_id=<ID>&text={text}` | B | free |
| **Home Assistant webhook** | `https://your-ha/api/webhook/xyz` | A | free if HA is running anyway |
| **Node-RED, n8n, your own service** | address of the flow | A or B | own server |
| **ntfy, self-hosted** | `https://ntfy.my-server/my-topic` | A | free, needs a server |
| **ntfy.sh (hosted)** | `https://ntfy.sh/my-topic` | A | limits on the free tier, plans above that |
| **IFTTT Webhooks** | `https://maker.ifttt.com/trigger/<event>/with/key/<key>?value1={text}` | B | free tier limited |

**Recommendation for the standard case: Telegram.** No server, no subscription, no further service to get to know – a bot with @BotFather, and the message lands in a chat. If you run Home Assistant, you are better off with the webhook: then the automation decides where the message goes, and can also serve services that the device cannot address itself.

**Pushover and the like do not work directly.** They require a POST request with several form fields; neither of the two modes fits that. The way there is through a webhook in Home Assistant or Node-RED – it receives the

message and passes it on.

## Setting up Telegram

**1. Create a bot.** In Telegram write to **@BotFather**, send `/newbot`, enter a display name (such as `CamperMinder`), then a unique user name ending in `bot`. BotFather replies with the **token**:

```
8123456789:AAHxyz-abcDEF_ghiJKL1234567890mnop
```

**2. Start the chat.** Open your own bot and press **Start**. A bot may not write to anyone who has not written to it first – without this step everything that follows fails.

**3. Get the chat ID.** Call up in the browser:

```
https://api.telegram.org/bot<TOKEN>/getUpdates
```

The reply contains `"chat":{"id":123456789,...}` – that number is the chat ID. If `{"ok":true,"result":[]}` comes back, step 2 is missing.

**4. Check it in the browser** before typing anything into the device:

```
https://api.telegram.org/bot<TOKEN>/sendMessage?chat_id=<NUMBER>&text=Test
```

`{"ok":true,…}` and the message on your phone means: it works. With `403` step 2 is missing, with `401` the token is wrong.

**5. Enter it in the device** – the same address, with the placeholder instead of `Test`:

```
https://api.telegram.org/bot<TOKEN>/sendMessage?chat_id=<NUMBER>&text={text}
```

*System → Remote alert*, enter it, **Save remote alert**, **Send a test message**. The address needs about 120 characters, the field holds 191.

## If you already have a bot

If you already use Telegram with Home Assistant or another system, you need **no new bot** – and do not have to change anything: **the device only sends, it never reads.** It calls only `sendMessage`. Several senders on the same bot do not get in each other's way. An existing webhook, or Home Assistant polling at the moment, is left untouched.

- **Token:** is in the configuration of the existing system (for Home Assistant under `telegram_bot: api_key`). Otherwise in Telegram with **@BotFather** → `/mybots` → choose the bot → *API Token*.
- **Chat ID:** likewise from the existing configuration (for Home Assistant `allowed_chat_ids`).

> **Do not call `getUpdates` in this case.** Telegram allows only one poller at a time: if a webhook is set for the bot, `getUpdates` answers with `409 Conflict`; if another system is polling right now, you snatch the message away from it. The chat ID therefore comes from the existing configuration – or from a second bot of your own.

**A bot of your own is still worthwhile** if the alarm message should sound different or land in a different chat from the rest of your home automation. Both work; it is a question of taste, not of technology.

### The token is a secret

> Whoever has the token can write as this bot. It is stored in the device and can be read on the device page – so, while the own network is open, by anyone within radio range. **If you use the remote alert, set a password under *System* → *Own network*.** And it does not belong on any screenshot.

### What the remote alert cannot do

- **It needs internet at the vehicle.** Without a Wi-Fi entered that has a connection to the outside, nothing goes out – on a pitch without reception all that remains is the buzzer and the message in the device.
- **It does not know whether the message arrives.** The device only learns from the service whether it accepted it. What happens after that is up to the service and your phone. That is why the test message.
- **One message per alarm.** The alarm stays on in the device until you confirm it with *Acknowledge alarm* – it is not sent repeatedly.

## 9 · Software updates

Updates bring new features and improvements to **LEVEL**. Nothing is installed until you confirm it.

| Situation | Way |
|---|---|
| Device has internet | Tab *System* → **Check for updates** |
| Device on its own network | Tab *System* → upload the update file from your phone |

The running version is shown in the same tab.

**Check for updates** asks GitHub and tells you what it found:

- **"You already have the latest version (…)"** – nothing to do, the device simply carries on running.
- **"New version … is being installed"** – the device downloads it, installs it and restarts. Afterwards **keep it powered for at least a minute**, then the new version counts.
- **"The device has no internet right now"** – the device is not on a Wi-Fi with internet, for example on the pitch. Then upload the file from your phone, see below.
- **"GitHub could not be reached"** – try again later.

Only a **higher** version is installed, never an older one.

> **You do not have to press anything for this.** If the device has internet, it checks by itself after every power-on and every 12 hours after that. If there is a new version, a blue notice appears at the top of the page, on every tab: *"New version … available. Tap here to install."* It is only installed when you tap and confirm – never on its own, so not while driving either.

### Uploading the update file from your phone

1. From the release on GitHub, download the file `level-firmware.ota.bin` to your phone. **Not the** `factory.bin`: that one is for a blank board, and the device rejects it as an update.

2. *System* → *Software* → *Choose file*, then **Upload file**. The page reports “Uploaded. The device is restarting.”
3. **Keep it powered for at least a minute.** Only then does the new version count as good. If power is lost before that, the device goes back to the old version the next time it is switched on – deliberately, so that a faulty update cannot cripple it.
4. Reload the page: under *System* → *Software* the new number is shown.

## 10 · With Home Assistant (optional extra)

As soon as the device is on your Wi-Fi, Home Assistant finds it by itself via ESPHome. The **CamperMinder** integration comes through HACS and brings the control card with it; Home Assistant then reports updates for the software in the device automatically.

> **The entity names of the device stay German.** Home Assistant shows the entities of the device under their German names, for example *Wächter Alarm*. The software in the device deliberately does not translate them, so that existing installations – automations, dashboards, MQTT – do not break when the language is switched. Only the control page itself changes language. The entities of the integration and the sentences in the card about the pitch (sun, satellite, Starlink) are also German at the moment; this chapter gives the translation where it matters.

Almost everything else is set in one place: *Settings* → *Devices & services* → *CamperMinder* → *Configure*. All fields there are optional; whatever stays empty simply does not exist.

### Messages to your phone: the blueprints

Five ready-made automations bring the messages to where you read them. In Home Assistant: *Settings* → *Automations & scenes* → *Blueprints* → *Import blueprint*, then paste the address:

- Fridge warning: `https://github.com/rcdev67/camperminder/blob/main/blueprints/automation/camperminder/kuehlschrank_warnung.yaml`
- Guard alarm: `https://github.com/rcdev67/camperminder/blob/main/blueprints/automation/camperminder/waechter_alarm.yaml`
- Frost warning: `https://github.com/rcdev67/camperminder/blob/main/blueprints/automation/camperminder/frostwarnung.yaml`
- Awning weather guard: `https://github.com/rcdev67/camperminder/blob/main/blueprints/automation/camperminder/markise_wetter.yaml`
- Departure check: `https://github.com/rcdev67/camperminder/blob/main/blueprints/automation/camperminder/abfahrt.yaml`

Then *Create automation* → *Use a blueprint*, choose the device and enter under “Was soll passieren?” (“What should happen?” – the blueprints are in German) how you want to be notified.

## Home location follows the vehicle

If Home Assistant runs in the vehicle, its home location usually still sits where it was set up. Sunrise and sunset, the weather and everything about the pitch depend on it. Under *Configure*:

- **Vehicle GPS source** – a tablet or phone that stays in the vehicle (the Home Assistant app reports its location). LEVEL itself has no GPS. With a source there is also the **"Set home here"** button.
- **Home location follows the vehicle** – when the vehicle has stood for five minutes after a drive at a place more than 500 m from the old one, that place becomes home. Only what LEVEL measures as a drive counts; taking the tablet to a restaurant moves nothing.

Only switch this on if Home Assistant runs in the vehicle. A Home Assistant at home keeps its home at home.

## Weather guard for the awning

Under *Configure*, choose a **weather forecast**, ideally Open-Meteo with the zone "Home" – it follows the home location and reports gusts. Every half hour the integration checks the next hours. For gusts, storm, thunderstorm, hail, snow or heavy rain the message is *retract the awning*; for a lot of rain it reminds you to lower the awning on one side. The limits are sliders on the *CamperMinder* device. The card shows the warning as a yellow bar; it reaches your phone with the *awning weather guard* blueprint.

If you also enter a sensor for the awning (**Awning extended**), you only get the warning while it really is out.

## The pitch: sun, satellite, Starlink

When you park, the direction of the vehicle decides three things: whether the awning is in the sun, whether the satellite dish has a clear view and whether Starlink gets open sky. For this, the bottom of the card shows a compass picture with three sentences below it.

**What the card shows.** North is always at the top; the vehicle turns, not the world:

- **Vehicle** in the middle, the arrow points to the front. Grey and dashed with "?" as long as the direction is missing.
- **Dashed yellow arc** – the path of the sun today, with times. The further inside, the higher the sun.
- **Yellow circle** – where the sun is right now.
- **Green line** – where the satellite dish has to point.
- **Blue area** – the sky Starlink needs to be clear.

Below are the sentences, to the degree, for example *"Astra 19,2° Ost: Südsüdost (164°), 30° hoch – vom Fahrzeug aus 17° links von vorne."* (south-south-east, 164°, 30° high – seen from the vehicle 17° left of straight ahead) or *"Starlink: Norden liegt 20° links von hinten"* (north lies 20° left of straight behind).

**Set once**, on the *CamperMinder* device: **Awning on side** (right or left in driving direction), **Fridge vents on side** (where the vent grilles are) and **Satellite** (Astra 19.2° E, Hotbird 13° E, Astra 23.5° E or Astra 28.2° E).

## The direction of the vehicle

All three sentences depend on where the front points. Below the compass picture it says where the direction came from and since when it applies.

**1. Automatically after every drive (recommended).** With a GPS source entered, CamperMinder takes the direction by itself one minute after you park. LEVEL measures how far the vehicle turns; the Home Assistant

app reports the driving direction while moving at speed. Together this gives the direction when parked – even if you turned round at the end and reversed in. The card then says "Aus der Fahrt übernommen" (taken from the drive). (With software in the device older than 4.6.0 there is only the driving direction; after reversing in, press **"↻ 180° drehen"** (turn by 180°) once.)

> **Important: switch on the high accuracy mode of the Home Assistant app.** Without it the app often reports its location only every few minutes while driving. Then the driving direction just before parking is missing, and no direction comes.

How to switch it on, on the phone or tablet that stays in the vehicle (Android):

1. Open the Home Assistant app, tap **Settings** at the bottom left.
2. Tap **Companion app**.
3. Tap **Manage sensors**.
4. In the *Location* section, open the **Background location** sensor.
5. Switch on **High accuracy mode** and set the interval below it to 5 seconds.

Depending on the app version the items may be named slightly differently. The mode uses more battery: if the device in the vehicle is plugged in, it can stay on. Otherwise, in the same place, it can be limited to the time the phone is connected to a particular Bluetooth device – such as the vehicle's radio. iPhone and iPad have no such mode; there the direction is better set with the phone compass or by hand.

**2. With the phone compass.** In the card press **" 📱 Handy-Kompass"** and hold the phone flat, top edge pointing in the driving direction. Inside the vehicle, metal often deflects the compass by 10° or more – it is more accurate beside the vehicle, phone flat in front of you, top edge parallel to the vehicle pointing forward. This only works over a secure connection (https, for example via Home Assistant Cloud).

**3. By hand.** Stand behind the vehicle with a compass app and look in the driving direction. Set the degrees with **−15° / −5° / −1° / +1° / +5° / +15°**: 0° is north, 90° east, 180° south, 270° west. The same is available as the **"Front points to"** slider on the *CamperMinder* device.

Set by hand or with the phone, the direction applies until the next drive. In the card, **"Wie stelle ich das ein?"** (how do I set this?) explains all three ways in short.

## Using the information when parking

> **Try it beforehand.** Before you park, you can turn the vehicle in the card with the buttons to the planned direction – the sentences immediately show what it would be like. Once the vehicle stands, the real direction follows by itself after a minute.

- **Sun.** The sentence names the times when the awning or the fridge vents are in the sun today. If you want shade under the awning in the afternoon, point it east or north; for warmth, rather south. The fridge vents belong in the shade – otherwise an absorber fridge cools noticeably worse on hot days.
- **Satellite.** *164°, 30° high* is the compass direction and the height above the horizon the dish has to point to. *17° left of straight ahead* means: seen from the driver's seat, a little left of straight on. Nothing may stand in exactly this direction and height – no tree, no building, no neighbouring vehicle. An automatic dish finds the satellite by itself but needs this clear view; a manual dish is set to the height, turned to the direction and then fine-tuned with the signal.

- **Starlink.** In the northern hemisphere the antenna aligns itself towards the north and needs a wide clear sky there. *North lies 20° left of straight behind* means: the antenna belongs where nothing is in the way towards the north – not in the shadow of your own body, not under trees. The obstruction check in the Starlink app shows the clear view exactly.

### Last time here: the pitch memory

With a GPS source, CamperMinder remembers at every pitch what it took to level the vehicle on arrival. If you come back to the same pitch, the top of the card shows for example "📍 *Hier am 11.10.2026: vorne links 2,5 cm (1 Stufe).*" (here on 11 Oct 2026: front left 2.5 cm, 1 step) – the wedges are in place before you get out. Anything within 60 m of last time counts as the same pitch. It is stored only in your own Home Assistant.

### Departure check

Awning out, step extended, roof vent open: **LEVEL** is the first to notice that the vehicle is rolling. Under *Configure*, choose in the field **"Departure check"** what has to be closed when driving off – door contacts, roof vents, step, locks; the awning of the weather guard is included automatically. If the vehicle starts rolling while something of it is still open, the card shows a red bar, **LEVEL** beeps, and with an announcement target it is announced. It reaches your phone with the *departure check* blueprint.

Both modes run with the same software in the device. If you start small, you can switch at any time – no new software, no new device.

## 11 · MQTT – connecting to other systems (optional extra)

The device reports tilt, lift height per corner and the instruction in plain language to an **MQTT broker**. That lets it be integrated into systems that have no tilt sensor of their own – such as ioBroker, openHAB or Node-RED.

> **Not needed for Home Assistant.** There everything runs through the built-in connection to the device. MQTT is the way to everything else. If you do not use MQTT, you will not notice it: with no address entered, the device does not open a connection.

> **Names and values stay German.** The topics below, the entity names and the text values they report (*aus*, *scharf in 45 s* …) are not translated by the software in the device, so that existing installations do not break when the language of the control page is switched. English meanings follow in brackets.

### Setting up

On the control page under **System** → **MQTT**: enter the broker address, port, user and password and save. After that the device restarts. The state is then shown in the **MQTT** field. **Clearing the address and saving switches MQTT off again.**

| Shown | Meaning |
|---|---|
| `MQTT is off` | no address entered – MQTT is switched off |
| `Connected to …` | running |

| | |
|---|---|
| `No connection to …` | Address reachable? Port? User and password? For this the device must be on the same Wi-Fi as the broker. |

## How the topics are named

```
camperminder/level/<art>/<name>/state
```

Whether the device is reachable is reported under `camperminder/level/status` with `online` or `offline`. Anyone processing values should listen to it – otherwise, after a power cut, they carry on calculating with the last known state. Signs as everywhere in the device: **lengthways plus = front higher, crossways plus = right side higher.**

## What you need for levelling

| **Topic (after `camperminder/level/`)** | **Content** |
|---|---|
| `sensor/neigung_pitch` | Lengthways tilt in degrees, + = front higher |
| `sensor/neigung_roll` | Crossways tilt in degrees, + = right higher |
| `sensor/hub_vorne_links`, `…_vorne_rechts`, `…_hinten_links`, `…_hinten_rechts` | Lift height per corner in cm |
| `sensor/stuetzrad` | Caravan: + up, − down |
| `sensor/abweichung_…` (per corner, and `abweichung_stuetzrad`) | Deviation in cm from fully level, also within the tolerance |
| `text_sensor/anweisung` | e.g. *Hinten links 4,5 cm hoch – Keilstufe 1* (“rear left 4.5 cm up – ramp step 1”) – the same sentence as on the control page |

On a **caravan**, “rear left/right” stand for the two wheels of the one axle; “front” stays at 0, that is where the jockey wheel is. The lift heights are **fully calculated** – vehicle dimensions, vehicle type and tolerance are already taken into account. The connected system does not have to calculate anything. **Lift or deviation?** The lift says what to do: if an axle is within the tolerance, it drops to 0. The deviation, on the other hand, always shows how far the corner is still from fully level.

## States

| **Topic (after `camperminder/level/`)** | **Content** |
|---|---|
| `binary_sensor/camper_steht_gerade` | `ON` = both axes level |
| `binary_sensor/pitch_eben`, `…/roll_eben` | `ON` = lengthways or crossways axis level |
| `binary_sensor/schraeglage` | `ON` = above the limit, default 3° (absorption fridge) |

| | |
|---|---|
| `binary_sensor/lageaenderung` | `ON` = vehicle has left its resting position |
| `binary_sensor/in_bewegung` | `ON` = shaking, i.e. someone present |
| `text_sensor/eigenes_netz` | `offen` (open) or `mit Passwort` (with password) |
| `text_sensor/firmware_version` | e.g. `4.2.4` |
| `text_sensor/update_stand` | Result of the update check: *prüft* (checking), *aktuell* (up to date), *Update verfügbar* (update available), *kein Internet* (no internet) … |
| `text_sensor/name_im_netz` | Name in the device list of the router, e.g. `camperminder-level` |

`in_bewegung` **is not an alarm.** It reports that someone is getting in, the wind is pushing, or the neighbour is manoeuvring. The alarm is `lageaenderung` – the vehicle has left its position.

## Self-monitoring

| **Topic (after `camperminder/level/`)** | **Content** |
|---|---|
| `binary_sensor/kalibrierung_pruefen` | `ON` = temperature or sensor values far from the calibration – calibrate again |
| `text_sensor/kalibrierung` | e.g. *am 22.08.2026 bei 21 °C* ("on 22.08.2026 at 21 °C"); *ab Werk kalibriert* ("factory calibrated") for as long as no calibration has been done in the vehicle |
| `binary_sensor/montage_pruefen` | `ON` = the sensor is delivering nonsense, housing loose? |
| `binary_sensor/frostgefahr` | `ON` = temperature below the limit (adjustable, default 3 °C) |
| `sensor/innentemperatur` | in °C. **Not a thermometer:** The chip heats itself up. Good enough for a frost warning, not for climate control. |

## Profiles (target profiles)

A simple spirit level knows one target: zero. **LEVEL** knows three profiles and levels accordingly: **Levelling**, **Sleeping** (head end a little higher) and **Draining** (boiler or tank drain towards one point). The profile affects *only* the levelling; the tilt warning and the guard carry on calculating with the real tilt.

| **Topic (after `camperminder/level/`)** | **Content** |
|---|---|
| `select/zielprofil` | `Ausrichten`, `Schlafen`, `Ablassen` (levelling, sleeping, draining) |

| | |
|---|---|
| `select/schlafrichtung` | `Kopf vorn`, `Kopf hinten`, `Kopf links`, `Kopf rechts` (head at the front, rear, left, right – seen in the direction of travel) |
| `number/kopfende_anheben` | 0 … 15 cm, default 2 |
| `select/ablasspunkt` | the *lowest* point: front, rear, left, right or one of the four corners |
| `number/neigung_zum_ablasspunkt` | 0 … 15 cm, default 5 |
| `sensor/ziel_laengs`, `sensor/ziel_quer` | the target tilt currently in force, in cm – the convenient way for a connected system |

## Fridge warning

`schraeglage` reports the **angle**. What damages an absorption fridge, however, is the angle **times the time** – being briefly tilted while manoeuvring is harmless. If you want to trigger a notification, therefore use `kuehlschrank_warnung`.

| **Topic (after `campermind er/level/`)** | **Content** |
|---|---|
| `binary_sensor/kuehlschrank_warnung` | `ON` = tilted for long enough that the fridge suffers |
| `sensor/schraeglage_dauer` | Minutes above the limit, `0` when not currently |
| `text_sensor/kuehlschrank` | *arbeitet normal* (“works normally”) or e.g. *3,4° schief seit 42 min – die Kühlleistung fällt ab* (“3.4° tilted for 42 min – cooling performance is dropping”) |
| `number/kuehlschrank_kritisch_nach` | 5 – 240 minutes |

Three levels from a single control: a notice as soon as the angle is exceeded, a warning after a third of the set time, urgent after the full time.

## Guard

`lageaenderung` goes off again as soon as the vehicle returns to its position. That is no use as an alarm – whoever is asleep misses it. The guard turns it into a state that **latches** and stays on until it is acknowledged. The alarm survives a power cut.

| **Topic (after `camper minder/level/`)** | **Content** |
|---|---|
| `binary_sensor/waechter_alarm` | `ON` = has triggered and is not acknowledged |
| `text_sensor/waechter` | *aus* (off), *scharf in 45 s* (armed in 45 s), *scharf seit 2 h 10 min* (armed for 2 h 10 min), *ALARM – Lage verändert am 22.08. um 03:14* (alarm – position changed on 22.08. at 03:14) |

| | |
|---|---|
| `text_sensor/letzte_bewegung` | *gerade jetzt* (right now), *22.08. um 21:40* (22.08. at 21:40), *vor 3 h 12 min* (3 h 12 min ago) |
| `switch/waechter` | `ON` / `OFF` – arm |
| `number/waechter_karenzzeit` | 0 – 600 s |
| `button/alarm_quittieren/command` | `PRESS` – clears the alarm **without** switching the guard off, and takes the current position as the new reference point |

If there was no valid clock when it triggered – the normal case on a pitch without internet – the text gives a time span instead of a time of day.

### Setting values via MQTT

These topics also accept commands ( `…/command` ):

| **Topic (after `camperminder/level/` )** | **Values** |
|---|---|
| `number/schraeglage_grenzwert` | 1 – 10 degrees |
| `number/lageaenderung_grenzwert` | 0.2 – 5 degrees |
| `number/waechter_karenzzeit` | 0 – 600 seconds |
| `switch/praezisionsmodus` , `switch/waechter` | `ON` / `OFF` |
| `switch/status_led` | `ON` / `OFF` – the light on the device |
| `switch/alarmton` | `ON` / `OFF` – the buzzer on alarm |
| `switch/auffahrton` | `ON` / `OFF` – the audible ramp guidance. Always comes back *off* after a restart and switches itself off when the target is reached. |
| `button/neigung_kalibrieren/command` | `PRESS` – take the current position as level |
| `button/kalibrierung_zuruecksetzen/command` | `PRESS` – delete the calibration in the vehicle; the factory calibration applies again |

The ramp guidance tone beeps faster the smaller the largest lift of any corner is – 1000 ms from ten centimetres, 120 ms at the target, then one long tone. It follows the **Target profile**. **A latched alarm flashes even when the Status LED is switched off** – the switch takes away the operating indicator, not the alert.

For levelling, `anweisung` and the four `hub_*` are enough – the rest is a bonus.

## 12 · If something does not work

| Problem | What helps |
|---|---|
| No Wi-Fi **CamperMinder** in the list | Is the green light on? If not, power is missing. If the red light flashes briefly every three seconds, the device is on your Wi-Fi, and its own network is therefore off: see *Joining your own Wi-Fi*. After switching on, wait up to 20 seconds. |
| The camera does not react to the codes | In the camera settings switch on scanning of QR codes, or use Google Lens. Otherwise use the ways under *Without the sticker*. |
| Code 1 is read, but the phone does not join the network | If the own network has been given a password (*System → Own network*), code 1 no longer fits: choose the network by hand and enter your password. |
| Nothing opens after code 1 | Normal on Android: scan code 2. On Samsung, *Wi-Fi → gear next to CamperMinder → Manage router* also works. |
| Code 2 opens nothing, or "Page not reachable" | The phone is not (or no longer) on the CamperMinder Wi-Fi. First code 1, then code 2 again. |
| Code 3 opens nothing | Code 3 only works if the device is on your Wi-Fi and the phone is, too. On some Android phones `.local` does not work at all: then use the icon from *Joining your own Wi-Fi*, step 3, or the address from the router. |
| The phone jumps out of the CamperMinder Wi-Fi | It is looking for internet and prefers your Wi-Fi. Confirm *Stay connected*; if that does not help, briefly switch off *Auto-connect* for your Wi-Fi, or the mobile data. |
| `192.168.4.1` does not load | Type it in full: `http://192.168.4.1`. Some browsers otherwise turn it into a search. The phone must be on the CamperMinder Wi-Fi. |
| Lengthways and crossways are swapped | The housing is installed turned. The arrow must point forwards. |
| The bubble jumps from edge to edge at the slightest movement | Installed upside down, but *Mounting* is set to *Lid facing up*. Set it to *Lid facing down* and calibrate again. |
| No longer reachable after entering the Wi-Fi | *If the page does not find the device*, section 7. |
| Back to square one | *System → Factory reset*. Deletes the Wi-Fi, the calibration in the vehicle and the vehicle dimensions; the factory calibration stays. A forgotten password for the own network is gone with it, too – nobody is locked out. |